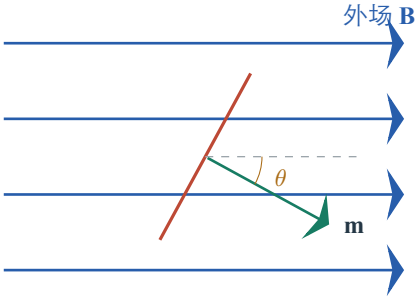
M5-153 转动导电圆环的非线性磁阻尼



红线为电流环侧视，**m** 垂直环面

$$S = \pi r^2,\ J = mr^2/2$$
$$\mathcal{E} = BS\omega\sin\theta$$

$$J\dot{\omega} = -B^2S^2\omega\sin^2\theta/R$$

$$\omega(\theta) = \omega_0 - \frac{B^2S^2}{JR}\left(\frac{\theta}{2} - \frac{\sin 2\theta}{4}\right)$$